

Extracting Latent Insights and Tagging Fall Injuries from Clinical Narratives using Unsupervised Learning

Naga V S M Rayabharam
Department of Computer Science
Pennsylvania State University
Harrisburg, USA

siddharthrayabharam.imp@gmail.com

Mohammad Masudur Rahman
School of Computing & Informatics
University of Louisiana at Lafayette
Louisiana, USA

mohammad.rahman3@louisiana.edu

Rahmanul Hoque
School of Business
University of the Cumberlands
Kentucky, USA

hoque.rahmanul@gmail.com

Sayed Mohsin Reza
Department of Computer Science
Pennsylvania State University
Harrisburg, USA
skr6024@psu.edu

Md Faisal Kabir
Department of Computer Science
Pennsylvania State University
Harrisburg, USA
mpk5904@psu.edu

Abstract—Falls have been one of the leading causes of death in the United States. Any insights about these falls can potentially inform interventions such as education and awareness programs, exercise programs, and home safety measures. As the accumulation of clinical text data by healthcare organizations is growing, it prompts a need for further insightful exploration into the application of machine learning techniques on these data. This paper discusses the use of a state-of-the-art language model to extract concepts from narratives. We propose novel unsupervised approaches such as Latent Dirichlet Allocation (LDA) and K-means to classify the narratives into multiple classes of injury type. In addition, we discuss a human-assisted survey method to evaluate the performance of the models. The proposed model demonstrated the classification accuracy of 83% (LDA) and 75% (K-means) on unlabeled clinical narratives. The approach yielded valuable insights that might not be readily apparent through human observation. Overall, this study demonstrated the feasibility of utilizing unsupervised NLP methods to extract information from clinical narratives.

Index Terms—Electronic Health Record, Fall Injury, Unsupervised Learning, Medical Narratives, Clinical Narratives, Topic modeling, Injury Classification

I. INTRODUCTION

Falls are one of the leading causes of injury-related death in older adults (≥ 65 years) [1]. Almost 165,005 older people died as a result of falls in the 2016-2020 time period; 261 million were treated in emergency rooms for falls-related injuries and nearly 4.4 million of these patients were hospitalized. In 2014, 28.7% of the elderly reported falling, resulting in an estimated 29.0 million injuries [2]. Any insights about these falls can potentially inform interventions such as education and awareness programs, exercise programs, and home safety measures.

An Electronic Health Record (EHR) is a digital version of the patient's paper chart. EHRs are patient-centered real-

time records that make information available instantly and contain patient medical history, diagnoses, medications, treatment plans, immunization dates, allergies, radiology images, and laboratory and test results [3]. Consumer Product Safety Commission (CPSC) has operated National Electronic Injury Surveillance System (NEISS) program that publicly hosts electronic health records of such fall-related injuries among older adults. NEISS collects data from emergency departments from multiple hospitals including significant inner-city trauma centers as well as large urban, suburban, rural, and children's hospitals. It uses a subset of these hospitals to obtain data related to medical narratives on injuries [4]. We explore the potential of employing machine learning techniques to analyze medical narratives, with the aim of uncovering insights that may elude human observation.

Artificial Intelligence (AI) and Machine Learning (ML) techniques have been used in healthcare for decades. These advanced AI/ML tools have now become an important aspect of patient care in many cases for the health care providers. These tools are capable of helping with case classification and diagnosis [5], enhance image scanning and segmentation [6], support decision-making, predict the risk of disease [7], [8] and in neuroimaging [9], determine the optimal sample size to test and utilize electronic records to eliminate database errors [10]. Although many studies focus on AI/ML applications in healthcare, they predominantly center around medical imaging for predictions and diagnoses, leaving the textual components, such as medical narratives and other EHR aspects, under explored.

Clinical narratives on Fall Injuries are rich source of potential insights about how, when, and why people fall. Few studies [11], [12] employ rule-based supervision methods to categorize clinical narratives, but they typically limit the

classification to only two categories. Given the complexity of the fall dataset, such as a binary classification approach is insufficient. Moreover, these studies do not investigate alternative statistical techniques for generating weak labels in the training data. Antunes et al. [13] and Moen et al. [14] focus on exploring NLP techniques to determine the similarity between two medical words or sentences. However, there is a noticeable gap in the literature regarding the utilization of these similarity measures for further data extraction from clinical narratives. We propose a novel topic modeling approach to classify the narratives into multiple classes of injury types. In addition, we also process these topics more to uncover hidden knowledge about falls in the United States. This knowledge may offer a mechanism leveraging unstructured clinical text data for characterization, and monitoring of primary care practices and systems.

In this paper, we discuss the current state-of-the-art research on medical narratives in section II. In the following section III, a detailed explanation of the publicly available dataset and proposed natural language processing and machine learning techniques to create new insights on the clinical narratives is given. We show the results of the proposed methodology in section IV.

II. LITERATURE REVIEW

The Health Information Technology for Economic and Clinical Health Act of the United States enforced in 2009 has fostered immense growth in the use of Electronic Health Record (EHR). The number of healthcare institutions with an EHR systems has increased from 17% to 22% in 2009-10 [15]. One of the most challenges faced by researchers when using EHR data is to use the large amount of detailed patient information embedded in clinical narratives. Clinical narratives are often unlabeled due to their large size and resource-intensive labeling requirements. In the past decade, many studies have been published using natural language processing (NLP) techniques to extract and model information embedded in such unlabeled clinical texts. While text classification is one of the sought-after application of clinical text mining, many studies focus on other applications like Named Entity Recognition (NER), also known as concept extraction, weak/distant supervision, word/sentence embeddings, topic modeling and clustering. We briefly discuss some of the mentioned techniques that are used in understanding and implementing the project.

An embedding is a mathematically meaningful vector representation of a word, a sentence, or a text that designed so that words or sentences with similar meanings or contexts have similar vectors. The similarity between two different embedding vectors is calculated using cosine similarity. In recent years, several word embedding models and pre-trained models [16], [17] which have been successfully applied in biomedical field. However, these vectors have two fundamental limitations that render them unsuitable for bio-medical fields. First, they use distinct vector to represent each word and do not consider the internal structure of the words. Second, these

models struggle for learning rare or out of vocabulary words present in the training data.

Zhang et al. [18] create a new word embedding model which outperforms current state-of-the-art embedding models in all bench-marking tests. The authors train the model on PubMed Central (PMC) corpora and use medical subject headings (MeSH), and Unified Medical Language System (UMLS) as bio-medical domain knowledge. They construct MeSH term graph based on its RDF data and then sample the MeSH term embeddings. The fastText subword embedding model is used to learn the distributed word embeddings based on text sequences and MeSH term sequences. From clinical data mining standpoint, such embeddings not only excel in representing the irregularities of the medical text but also can incorporate structured information beyond what is found in text, proving beneficial for NER. NER is the task of identifying and locating mentions of conceptual category, in clinical texts, these are drug, symptom, or disease names. It is one of the most common extraction task that is run on the clinical texts. Tulkens et al. [19] improve this strategy by proposing a composition function for word representations of text and UMLS dictionary terms and subsequently evaluating using cosine similarity. Stanza [20] offered state-of-the-art NER pipeline which are trained on biomedical [21], [22] and clinical domain database [23]. The authors train a forward and a backward LSTM character-level language model (CharLM) to get the word representation in each sentence which feeds a BiLSTM [24] POS tagger with conditional random decoder. We also explore spaCy's [25] NER model which is trained on BC5CDR dataset [22] to accomplish our NER tasks. One of the major advantage of NER systems is their flexibility in learning to tag different types of entities by swapping training datasets. However, clinical NER has the issue of domain specificity. The clinical narratives are complex and contain specialized medical terms, numerical scores and abbreviations which makes NER suffer from reduced accuracy.

Many machine learning approaches to clinical NLP typically face challenges stemming from the scarcity of structured training data as most of the clinical texts are unstructured and unlabeled. Wang et al. [11] and Yao et al. [12] proposed a solution using keyword-based weak labels to classify the texts and then deep learning models to classify the clinical narratives. Clustering algorithms like Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) [26] has gained attention in recent years due to its ability to uncover complex and irregularly shaped clusters within data. Topic modeling techniques like Latent Dirichlet Allocation (LDA) [27], Latent Semantic Indexing (LSI) [28] and variants of Fuzzy Latent Semantic Analysis (FLSA) [29] are recently gaining attention due to its adaptability to various types of texts and ability to discover hidden patterns in the documents by representing the documents in terms of topics.

While existing research emphasizes the application of unsupervised NLP techniques, particularly focusing on word embeddings and named entity recognition in clinical narratives, there remains a notable gap in comprehending the distinctive

advantages that alternative unsupervised techniques, like topic modeling, might offer. Addressing this gap is pivotal for advancing our understanding of the latent structures within clinical narratives, thereby contributing to more nuanced and effective classification methodologies. This paper seeks to fill this gap by exploring the potential of topic modeling and K-means in classifying clinical narratives for extracting insights and identifying fall injuries events.

III. METHODOLOGY

This section presents a detailed overview of the method and dataset employed in the study. We further elaborate the preprocessing steps to clean the dataset. Finally, we explore different unsupervised NLP techniques applied to extract hidden insights from the clinical narratives and classify them.

A. Dataset

The study utilizes a large public dataset known as the National Electronic Injury Surveillance System (NEISS), which is managed and maintained by the Consumer Product Safety Commission. The dataset includes around 115,000 confirmed cases of unintentional falls among older adults. It comprises various attributes such as demographic information of the patients, incident descriptions, treatment dates, clinical diagnoses, and affected body parts of the patients, and a sample narrative is shown in Figure 2. These attributes are utilized as features for our experiment.

B. Preprocessing

The narratives contain many medical abbreviations that are widely used by the medical professionals to improve the efficiency without possibilities of obfuscation and ambiguity. A curated list of medical abbreviations relevant to the dataset was created from Wikipedia [30] as shown in Table I. The abbreviations are then translated into their full form in order to improve the context in the narratives. The overview of the pre-processing steps applied to achieve the improved and processed narratives is in Figure 1. We also translate numerically encoded columns into textual columns. After translation, we adopt the method proposed by Hegselmann et al. [31] to transform tabular data into fully textual data. To achieve this, we created a standard template of text such that all other important columns are accommodated along with the clinical narrative to create a descriptive narrative as shown in Figure 2. We will refer to this descriptive narrative as “detailed narrative” from hereafter in the paper. To improve the quality of the data we explore a pre-trained text-to-text transformer, FLaan-PaLM-540B, to extract particular incident related information. By using the information from other textual columns and feature values, detailed narratives often enable effective *zero-shot* or *few-shot* learning LLM (Large Language Model) models.

As proposed by Hegselmann et al. [31], we provide these detailed narratives to Google T5 model, a pre-trained text-to-text transformer. However, the challenge is to ask questions that are general enough for the model to return answers that

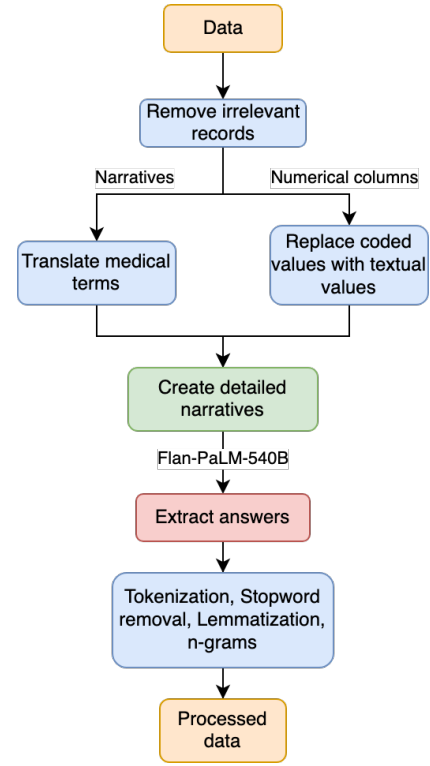


Figure 1: Overview of the pre-processing steps applied to the medical narratives

align with the information we’re intended to extract from the narratives like *action*, *cause*, *diagnosis*, *how*, *what*, and *where*. We have specifically narrowed down the questions to the following small set as following:

- What was the patient doing at the time of the incident? (*action*)
- What caused the patient to fall? (*cause*)
- What is the full diagnosis? (*diagnosis*)
- How did the fall occur? (*how*)
- What happened to the patient during the incident (*what*)
- Where did the fall occur? (*where*)

The T5 transformer model provides answers to these questions, each with a length of up to 20 words. The answers to the questions are saved under separate column. One of caveats of this approach, is that some of these questions return similar or even same answers for certain narratives. However, it is better to have extra data than too few data. Another approach to avoid

Table I: List of few medical abbreviations and their full forms that will be used to replace terms in narratives.

Abbreviation	Full form
<i>dx</i>	<i>diagnosis</i>
<i>fx</i>	<i>fracture</i>
<i>pt</i>	<i>patient</i>
<i>biba</i>	<i>brought in by ambulance</i>
<i>bal</i>	<i>blood alcohol level</i>
<i>ams</i>	<i>altered mental status</i>
<i>tr</i>	<i>trauma</i>

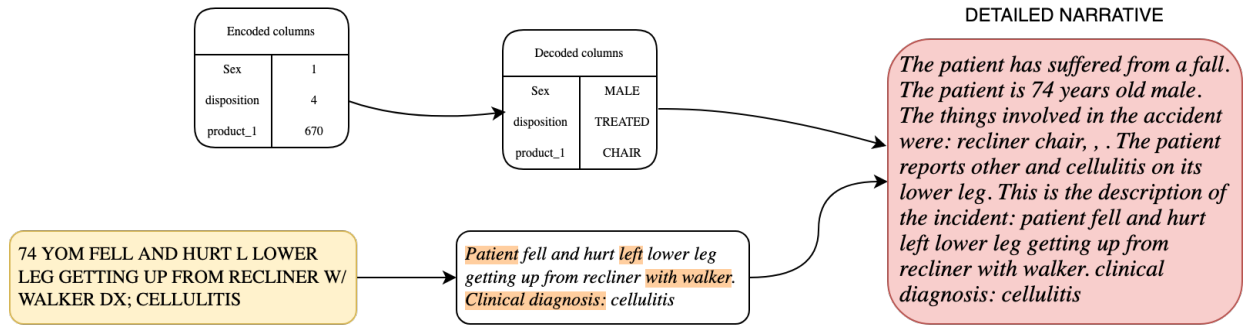


Figure 2: Overview of obtaining detailed narratives. We first translate the narratives to remove any medical abbreviations and then we decode the numerical columns. We combine translated narratives and decoded columns to create detailed narratives.

this issue is to train a model to generate questions that return specific answers but this would get computationally really expensive. So, we end up with above set of questions after improving them over multiple iterations after analyzing the answers after each iteration. The novelty in this approach lies in the ability of transformer model to extract information that might have been challenging due to the unstructured nature of clinical narratives.

The extracted answers are passed through a pipeline to enhance the quality of text representation for further tasks. The pipeline divides the sentences into individual words or subwords and eliminates any commonly occurring words like *the*, *is*, *it* etc. that do not contribute to the overall meaning of the text. These tokens are used to generate bigrams and trigrams capture contextual information and relationships between words. Finally, these words are lemmatized to ensure the consistency in the analysis. The output of this pipeline is a list of tokens for each answer which will be used for topic modeling and classification of clinical narratives.

C. Topic Modeling

We use Latent Dirichlet Allocation (LDA) model which is an unsupervised probabilistic generative model of topics. The basic idea is that documents are represented as random mixtures over latent topics, where each topic is characterized by a distribution over words. According to Blei et al. [27], A k -dimensional Dirichlet random variable θ takes the Dirichlet distribution. Given the parameters α (mixture of topics) and β (mixture of words), the joint distribution of a topic mixture θ , a set of N topics z , and a set of N words w is given by:

$$p(\theta, z, w | \alpha, \beta) = p(\theta, \alpha) \prod_{n=1}^N p(z_n | \theta) p(w_n | z_n, \beta)$$

where $p(z_n | \theta)$ is simply θ_i for the unique i such that $z_n^i = 1$. Integrating over θ and summing over z , we obtain the marginal distribution of a document:

$$p(w | \alpha, \beta) = \int p(\theta, \alpha) \left(\prod_{n=1}^N \sum_{z_n} p(z_n | \theta) p(w_n | z_n, \beta) \right) d\theta$$

Finally, taking the product of the marginal probabilities of single documents, we obtain the probability of a corpus:

$$p(D | \alpha, \beta) = \prod_{d=1}^M \int p(\theta_d, \alpha) \Gamma_d d\theta_d$$

where $\Gamma_d = \prod_{n=1}^N \sum_{z_{dn}} p(z_{dn} | \theta_d) p(w_{dn} | z_{dn}, \beta)$

Since the Dirichlet Distribution [32] is a continuous conditional probability distribution, Gibbs sampling [33] is applied on the distribution for digital computation. Gensim [34] is a python framework that provides easy-to-use implementation of Latent Dirichlet Allocation for topic modeling. We simply use gensim to create and train the LDA model on the answers that were obtained from T5 transformer model. For each question column, we create a term-document matrix which represents the frequency of terms (words or phrases) in documents in a collection. This term-document matrix will be the input of the LDA model. Apart from the term-document matrix, the LDA model also takes a lot of hyperparameters like number of topics, trigram and bigram count, chunksize and passes. We run Optuna [35] framework on a ranges of values create an optimal training model. Table II shows optimal hyperparameters for each column.

D. Clustering

We also evaluate the potential of several clustering models as a baseline to assess the performance of LDA. Specifically, we select K-means which offers us the flexibility of choosing number of clusters needed. DBSCAN and HDBSCAN variants change the number of clusters based on the data

Table II: Hyperparameters for Latent Dirichlet Allocation (LDA) obtained after going through 40 trails per column using Optuna framework. They are optimized on 30000 rows.

Column	Num Topics	Chunk size	passes	Bigram count	Trigram count
<i>action</i>	10	1500	4	1	100
<i>cause</i>	10	1000	1	4	70
<i>diagnosis</i>	10	500	1	1	100
<i>how</i>	70	500	4	4	10
<i>what</i>	60	500	1	7	70
<i>where</i>	10	1500	1	4	10

distribution and other hyperparameters. For clustering, we use word2vec [36] to convert the answer tokens to vector representations. The word2vec model takes many parameters like vector size, number of windows, skip-grams etc. We optimize the hyperparameters by running a parameter grid search evaluated on the similarity of two very close words. The word2vec model is trained on the entire corpus of answer tokens including the keyword tokens which will be discussed in section III-E. Now, we combine the tokens from ‘diagnosis’ question with keyword tokens and provide all these tokens to the clustering model to form 8 clusters, ideally, one for each class. The hyperparameters of the models are also optimized using Optuna based on Silhouette Score, Calinski-Harabasz Index and Davies-Buldin Index [35].

E. Classification

We classify the narratives into 8 different injury types which cover most common injury types. Each class is assigned with a keyword as specified by Hedegaard et al. [37] in their Center for Disease Control (CDC) publication. Table III provide the keywords used for each injury class.

These keywords are then added to the training corpus of all the models as different case file. All the narratives which have been grouped in the same cluster as these injury keywords are labelled with injury class associated with the keyword. A visual representation of the classification pipeline is depicted in Figure 3.

F. Evaluation

The evaluation of the models depends on the evaluation goals, which are operationalized through various metrics. The performance of LDA model is evaluated based on the perplexity and coherence metric. Perplexity captures how surprised a model is of new data it has not seen before, and is measured as the normalized log-likelihood of a held-out test set. On the other hand, coherence measures score a single topic by measuring the degree of semantic similarity between high scoring words in the topic. Similarly, for the clustering models we use Silhouette Score, Calinski-Harabasz Index and Davies-Buldin Index. Silhouette Score measures how well-separated clusters are. Calinski-Harabasz Index measures the ratio of between-cluster variance to within-cluster variance. Davies-Bouldin Index measures compactness and separation between clusters.

The optimal method for evaluating the classification results of an unlabeled dataset is through human assessment of model

Table III: Each class gets a particular keyword(s).

Injury class	Keyword(s)
0	fracture
1	sprain
2	strain
3	laceration
4	contusion, concussion
5	dislocation
6	hematoma, hemorrhage
7	injury, other

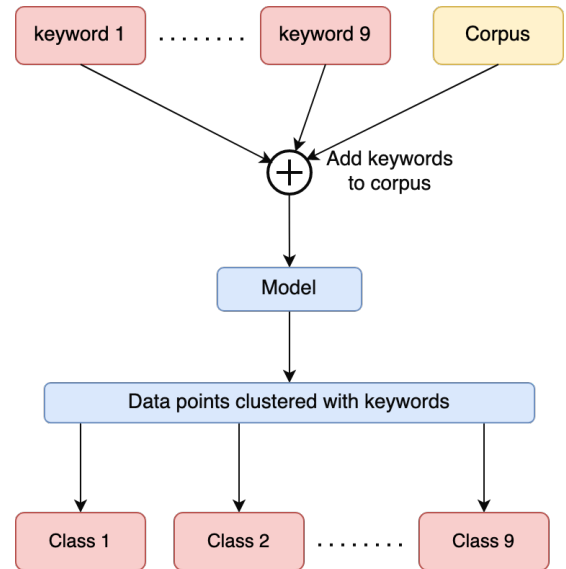


Figure 3: Overview of classification step. Every injury class is associated with a distinct keyword, recommended by the CDC [37]. These keywords are incorporated into the training corpus. Data points clustered or grouped with these keywords are classified as same injury type as the keywords.

performance. We refer the medical professionals as human in this regard. We created a test sample of 200 narratives, where each of the 20 participants was given 10 clinical narratives and asked to assign the appropriate injury class. We then compared the model predictions with the human evaluations to assess model performance.

IV. RESULTS

We subsequently compare the model predictions with the human evaluation results, offering insights into the models’ performance. Table IV presents a comprehensive breakdown of the class-wise accuracy achieved by both models in categorizing injury classes within medical narratives. K-means exhibits an overall accuracy of 72%, while LDA outperforms with an accuracy of 81%.

Figure 4 presents a comprehensive comparison of the class-wise F1 scores obtained by both models. Notably, both models exhibit sub-optimal performance in the classification of class

Table IV: Class-wise accuracy provided by K-Means and LDA models

Class	Accuracy	
	K-means	LDA
0	0.87	0.93
1	0.63	0.67
2	0.75	0.87
3	0.59	0.75
4	0.78	0.83
5	0.77	0.81
6	0.67	0.87
7	0.73	0.78
Overall	0.72	0.81

1, specifically related to sprains. A plausible explanation for the lower F1 score in this class could be attributed to the inadequacy of data size. The insufficient representation of instances belonging to class 1 in the dataset might impede the model to capture and generalize patterns effectively, leading to reduced performance of the model. In general, LDA outperforms the K-means clustering algorithm.

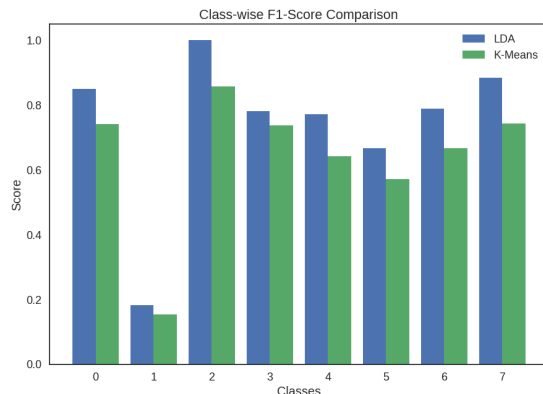


Figure 4: Plot of class-wise F1 scores.

Figure 5 illustrates the confusion matrix for these models. An interesting pattern emerges wherein clinical narratives belonging to fracture cases are consistently misclassified as other types of injuries in both models. This misclassification suggests the presence of substantial irrelevant data in the dataset. Consequently, further data preprocessing is necessary to enhance the model performance by eliminating this irrelevant data, ensuring more accurate and reliable injury classification.

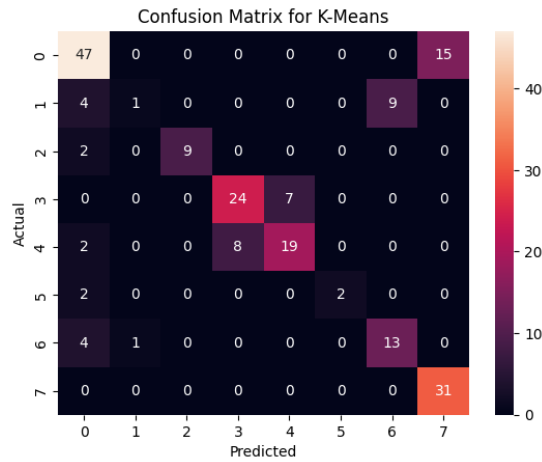
V. CONCLUSION

In this study, we present a novel approach of classifying clinical narratives into different injury types and improving insights on the data. We discuss in detail unique strengths and weaknesses owing to their underlying statistical properties of unsupervised learning techniques like Latent Dirichlet Allocation (LDA) and K-means paired with word2vec to tag or classify the clinical narratives and methods to evaluate the performance. As the adoption of EHRs increases and healthcare organizations accumulate larger volumes of clinical text data, such insights may provide a mechanism for leveraging unstructured clinical text data to characterize and monitor primary care practices and systems. Additionally, these insights can potentially inform interventions such as education and awareness programs, exercise and safety programs.

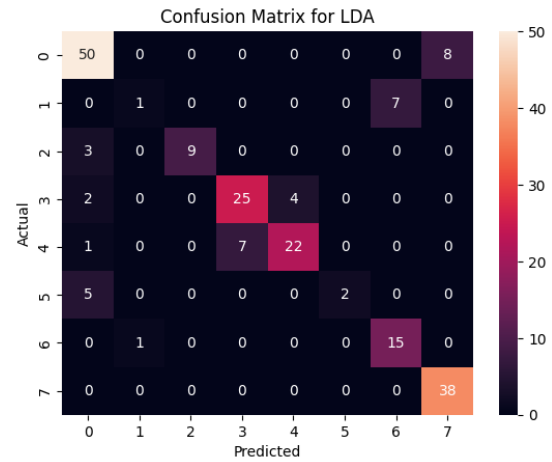
Despite notable challenges in handling specific injury types, future efforts should concentrate on refining or finding models to ensure generalized learning across all classes. Addressing limitations in certain injury types is crucial for the robustness of the classification system. Moreover, exploring automated evaluation mechanisms using advanced language models like GPT, BERT, or similar technologies holds promise for enhancing the assessment process.

REFERENCES

- [1] N. G. Choi, B. Y. Choi, D. M. DiNitto, C. N. Marti, and M. E. Kunik, "Fall-related emergency department visits and hospitalizations among community-dwelling older adults: examination of health problems and injury characteristics," *BMC geriatrics*, vol. 19, pp. 1–10, 2019.
- [2] Falls and fall injuries among adults aged above 65 years - united states, 2014. [Online]. Available: <https://www.cdc.gov/injury/features/older-adult-falls/index.html>
- [3] H. Zheng and S. Jiang, "Frequent and diverse use of electronic health records in the united states: A trend analysis of national surveys." *Digit Health*, vol. 8, p. 20552076221112840, Jan-Dec 2022.
- [4] National electronic injury surveillance system - all injury program (neiss-aip). [Online]. Available: <https://health.gov/healthypeople/objectives-and-data/data-sources-and-methods/data-sources/national-electronic-injury-surveillance-system-all-injury-program-neiss-aip>
- [5] Z. Liang, G. Zhang, J. X. Huang, and Q. V. Hu, "Deep learning for healthcare decision making with emrs," in *2014 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, 2014, pp. 556–559.
- [6] A. Esteva, B. Kuprel, R. A. Novoa, J. Ko, S. M. Swetter, H. M. Blau, and S. Thrun, "Dermatologist-level classification of skin cancer with deep neural networks," *Nature*, vol. 542, no. 7639, pp. 115–118, 2017. [Online]. Available: <https://doi.org/10.1038/nature21056>
- [7] B. P. Nguyen, H. N. Pham, H. Tran, N. Nghiem, Q. H. Nguyen, T. T. Do, C. T. Tran, and C. R. Simpson, "Predicting the onset of type 2 diabetes using wide and deep learning with electronic health records," *Computer Methods and Programs in Biomedicine*, vol. 182, p. 105055, 2019. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S016926071930327X>
- [8] Y. Ge, Q. Wang, L. Wang, H. Wu, C. Peng, J. Wang, Y. Xu, G. Xiong, Y. Zhang, and Y. Yi, "Predicting post-stroke pneumonia using deep neural network approaches," *International Journal of Medical Informatics*, vol. 132, p. 103986, 2019. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1386505618312723>
- [9] M. Faturrahman, I. Wasito, N. Hanifah, and R. Mufidah, "Structural mri classification for alzheimer's disease detection using deep belief network," in *2017 11th International Conference on Information and Communication Technology and System (ICTS)*, 2017, pp. 37–42.
- [10] C. Rudin and B. Ustun, "Optimized scoring systems: Toward trust in machine learning for healthcare and criminal justice," *Interfaces*, vol. 48, 10 2018.
- [11] Y. Wang, S. Sohn, S. Liu, F. Shen, L. Wang, E. J. Atkinson, S. Amin, and H. Liu, "A clinical text classification paradigm using weak supervision and deep representation," *BMC Med Inform Decis Mak*, vol. 19, no. 1, p. 1, Jan 2019.
- [12] L. Yao, C. Mao, and Y. Luo, "Clinical text classification with rule-based features and knowledge-guided convolutional neural networks," *BMC Med Inform Decis Mak*, vol. 19, no. Suppl 3, p. 71, Apr 2019.
- [13] R. Antunes, J. a. F. Silva, and S. Matos, "Evaluating semantic textual similarity in clinical sentences using deep learning and sentence embeddings," in *Proceedings of the 35th Annual ACM Symposium on Applied Computing*, ser. SAC '20. New York, NY, USA: Association for Computing Machinery, 2020, pp. 662–669. [Online]. Available: <https://doi.org/10.1145/3341105.3373987>
- [14] H. Moen, L.-M. Peltonen, M. Koivumäki, H. Suhonen, T. Salakoski, F. Ginter, and S. Salanterä, "Improving layman readability of clinical narratives with unsupervised synonym replacement," *Stud Health Technol Inform*, vol. 247, pp. 725–729, 2018.
- [15] W. H. Henricks, "“meaningful use” of electronic health records and its relevance to laboratories and pathologists," *J Pathol Inform*, vol. 2, p. 7, Feb 2011.
- [16] T. Mikolov, I. Sutskever, K. Chen, G. S. Corrado, and J. Dean, "Distributed representations of words and phrases and their compositionality," in *Advances in Neural Information Processing Systems*, C. Burges, L. Bottou, M. Welling, Z. Ghahramani, and K. Weinberger, Eds., vol. 26. Curran Associates, Inc., 2013. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2013/file/9aa42b31882ec039965f3c4923ce901b-Paper.pdf
- [17] J. Pennington, R. Socher, and C. Manning, "GloVe: Global vectors for word representation," in *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Doha, Qatar: Association for Computational Linguistics, Oct. 2014, pp. 1532–1543. [Online]. Available: <https://aclanthology.org/D14-1162>



(a) Confusion matrix for K-means clustering algorithm



(b) Confusion matrix for Latent Dirichlet Allocation algorithm

Figure 5: Confusion matrices

- [18] Y. Zhang, Q. Chen, Z. Yang, H. Lin, and Z. Lu, "Biowordvec, improving biomedical word embeddings with subword information and mesh," *Scientific Data*, vol. 6, no. 1, p. 52, 2019. [Online]. Available: <https://doi.org/10.1038/s41597-019-0055-0>
- [19] S. Tulkens, S. Šuster, and W. Daelemans, "Unsupervised concept extraction from clinical text through semantic composition," *Journal of Biomedical Informatics*, vol. 91, p. 103120, 2019. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1532046419300383>
- [20] Y. Zhang, Y. Zhang, P. Qi, C. D. Manning, and C. P. Langlotz, "Biomedical and clinical English model packages for the Stanza Python NLP library," *Journal of the American Medical Informatics Association*, 06 2021.
- [21] S. Pyysalo and S. Ananiadou, "Anatomical entity mention recognition at literature scale," *Bioinformatics*, vol. 30, no. 6, pp. 868–875, Mar 2014.
- [22] J. Li, Y. Sun, R. J. Johnson, D. Sciaky, C.-H. Wei, R. Leaman, A. P. Davis, C. J. Mattingly, T. C. Wieggers, and Z. Lu, "Biocreative v cdr task corpus: a resource for chemical disease relation extraction," *Database (Oxford)*, vol. 2016, 2016.
- [23] Ö. Uzuner, B. R. South, S. Shen, and S. L. DuVall, "2010 i2b2/va challenge on concepts, assertions, and relations in clinical text," *J Am Med Inform Assoc*, vol. 18, no. 5, pp. 552–556, Sep-Oct 2011.
- [24] Z. Huang, W. Xu, and K. Yu, "Bidirectional LSTM-CRF models for sequence tagging," *CoRR*, vol. abs/1508.01991, 2015. [Online]. Available: <http://arxiv.org/abs/1508.01991>
- [25] M. Honnibal, I. Montani, S. Van Landeghem, and A. Boyd. (2020) spaCy: Industrial-strength Natural Language Processing in Python.
- [26] C. Malzer and M. Baum, "Hdbscan: An alternative cluster extraction method for HDBSCAN," *CoRR*, vol. abs/1911.02282, 2019. [Online]. Available: <http://arxiv.org/abs/1911.02282>
- [27] D. M. Blei, A. Y. Ng, and M. I. Jordan, "Latent dirichlet allocation," *J. Mach. Learn. Res.*, vol. 3, no. null, pp. 993–1022, mar 2003.
- [28] M. A. Ahmad, C. Eckert, and A. Teredesai, "Interpretable machine learning in healthcare," in *Proceedings of the 2018 ACM international conference on bioinformatics, computational biology, and health informatics*, 2018, pp. 559–560.
- [29] E. Rijcken, F. Scheepers, P. Mosteiro, K. Zervanou, M. Spruit, and U. Kaymak, "A comparative study of fuzzy topic models and lda in terms of interpretability," in *2021 IEEE Symposium Series on Computational Intelligence (SSCI)*, 2021, pp. 1–8.
- [30] Wikipedia contributors, "List of medical abbreviations — Wikipedia, the free encyclopedia," https://en.wikipedia.org/w/index.php?title=List_of_medical_abbreviations&oldid=1177253313, 2023, [Online; accessed 10-November-2023].
- [31] S. Hegselmann, A. Buendia, H. Lang, M. Agrawal, X. Jiang, and D. Sontag, "Tabllm: Few-shot classification of tabular data with large language models," 2023.
- [32] Wikipedia contributors, "Dirichlet distribution — Wikipedia, the free encyclopedia," https://en.wikipedia.org/w/index.php?title=Dirichlet_distribution&oldid=1170494520, 2023, [Online; accessed 11-November-2023].
- [33] L. Martino, V. Elvira, and G. Camps-Valls, "The recycling gibbs sampler for efficient learning," *Digital Signal Processing*, vol. 74, pp. 1–13, 2018.
- [34] R. Rehurek and P. Sojka, "Gensim—python framework for vector space modelling," *NLP Centre, Faculty of Informatics, Masaryk University, Brno, Czech Republic*, vol. 3, no. 2, 2011.
- [35] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: A next-generation hyperparameter optimization framework," 2019.
- [36] T. Mikolov, K. Chen, G. Corrado, and J. Dean, "Efficient estimation of word representations in vector space," 2013.
- [37] H. Hedegaard, R. L. Johnson, M. F. Garnett, and K. E. Thomas, "The 2020 international classification of diseases, 10th revision, clinical modification injury diagnosis framework for categorizing injuries by body region and nature of injury," *Natl Health Stat Report*, no. 150, pp. 1–27, Dec 2020.